Uncertainty-Aware Deformable DETR

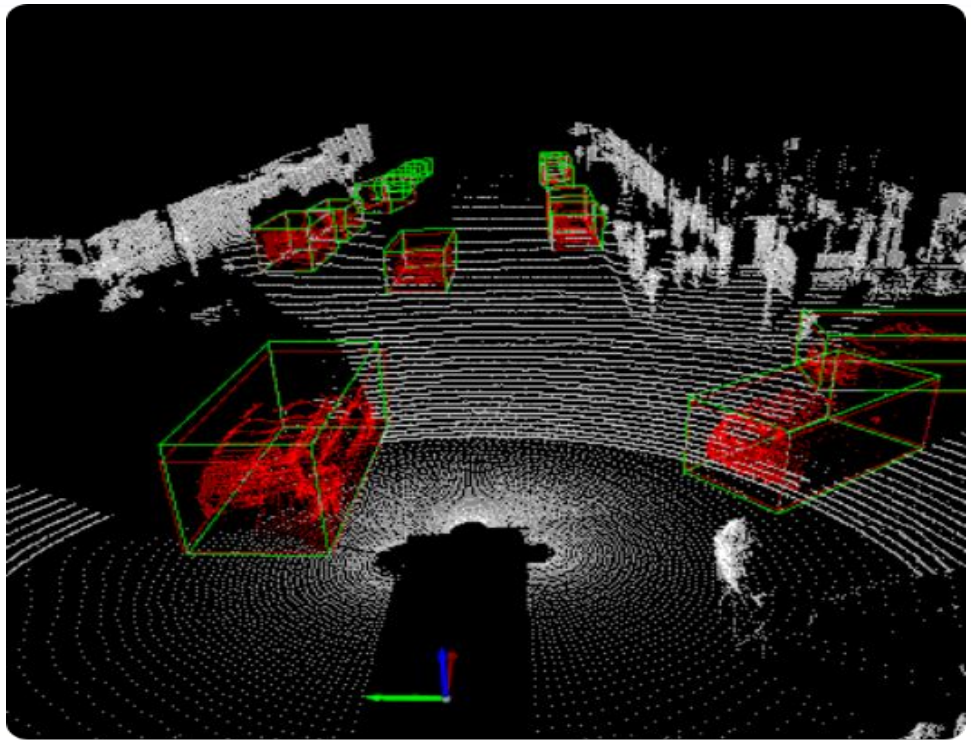
Calibrated Reliability for Next-Generation Object Detection Systems.

Problem Statement: Why this matters?

The Dangers of Overconfidence

Standard Transformer detectors are often **miscalibrated**. They report high confidence scores for incorrect bounding boxes.

- Failure to signal localization errors.
- Blindness to sensor degradation.
- Potential for catastrophic failure in autonomous driving or medical imaging.



Project Summary

The Core Mission

Our work evolves **Deformable DETR** from a simple "detection-only" model into an **uncertainty-aware** detector. We move beyond raw Average Precision (AP) to focus on **Reliability**.

> Retain detection strength while adding explicit confidence calibration and variance estimation.

- Improved Confidence Calibration
- Explicit Uncertainty Quantification
- Selective Decision Filtering

System Capabilities

The baseline model provides only a single point estimate. Our model distinguishes between:

Aleatoric: Is the data noisy, blurry, or occluded?

Epistemic: Is the model ignorant of this specific object or domain?

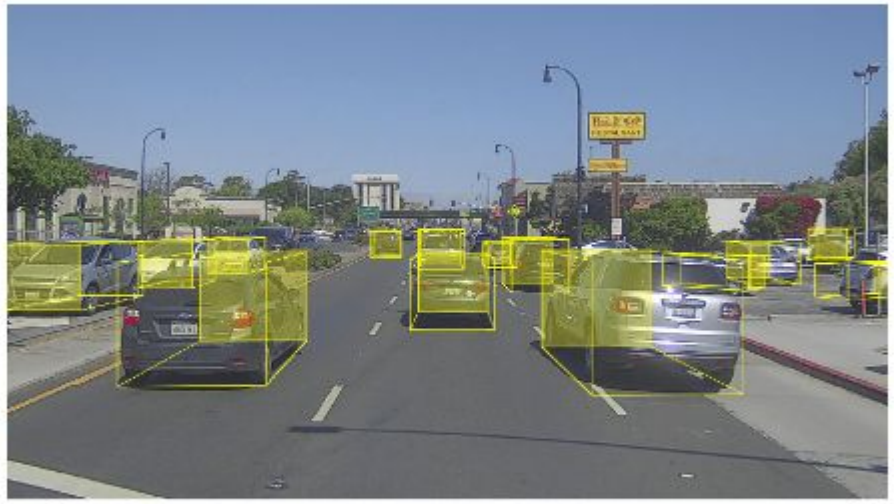
Track A: Aleatoric Variance Head

Models inherent noise in input data (e.g., sensor blur or occlusion).

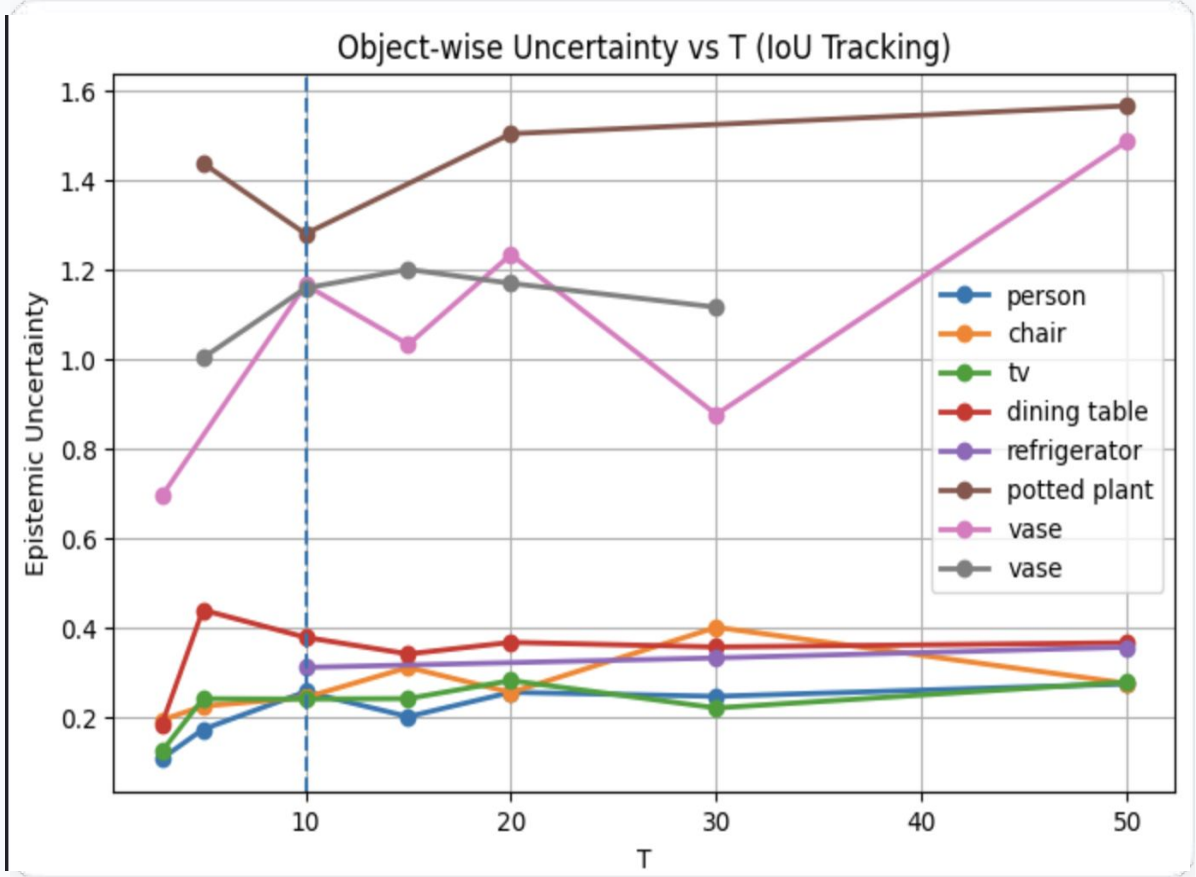
> Parallel Head predicts log-variance for each coordinate.

$$L = \frac{1}{2}e^{-s}\|y - \hat{y}\|_2^2 + \frac{1}{2}s$$

- Loss Attenuation allows the model to "explain away" noisy labels.
- Signals precision of bounding box localization.
- Learns per-box predictive variance in a data-driven way.



Track B: Epistemic MC Dropout

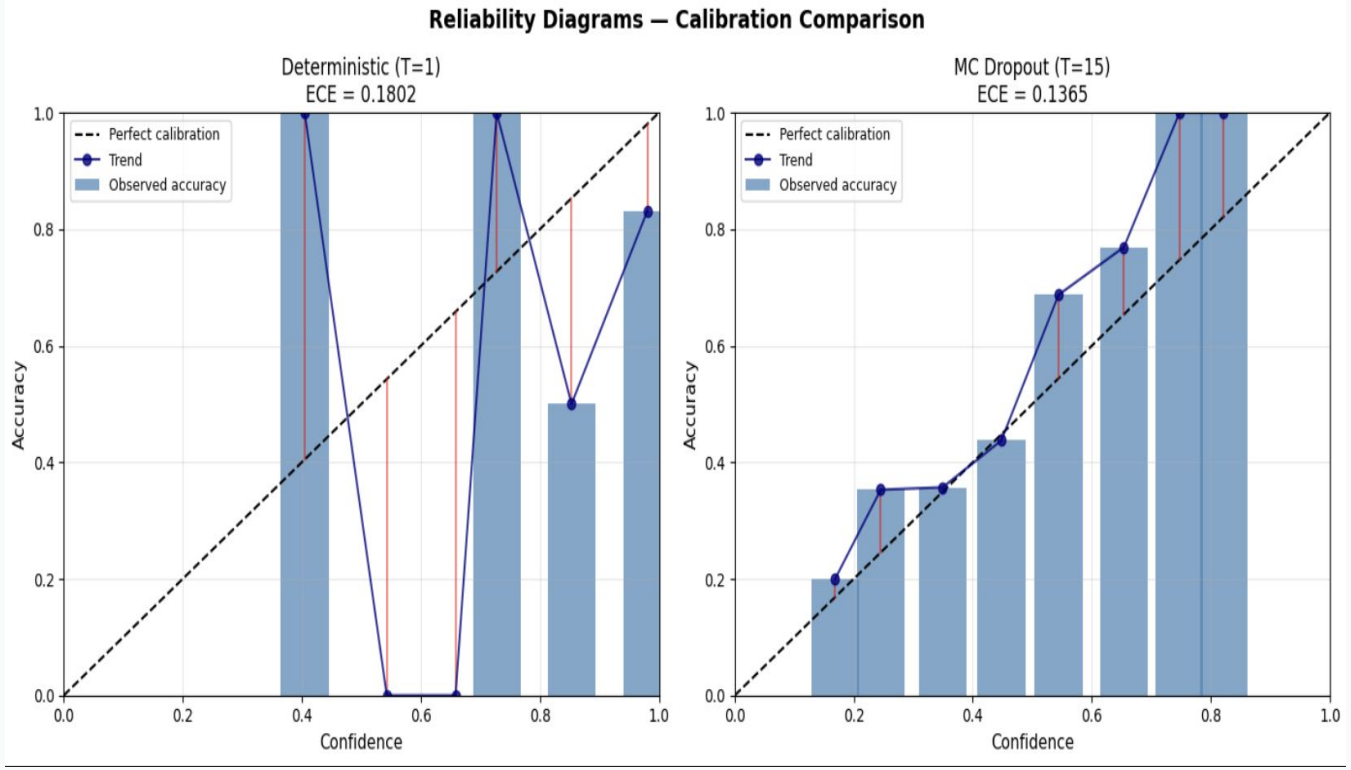


- We estimate epistemic uncertainty using MC Dropout .
- Perform multiple stochastic forward passes (T times)
- Compute variance across predictions → uncertainty
- We analyze uncertainty at object-level for different T

Observation:
Uncertainty stabilizes after T ≈ 15

Final Choice:
T = 15 (best trade-off between stability and computation)

Improving Model Calibration using Uncertainty



Problem:
- Deep models are often overconfident

Solution:
- Reduce confidence using uncertainty adjusted confidence

Evaluation:
- Expected Calibration Error (ECE)

Results:
- ECE reduced
- Predictions align better with actual accuracy
- Model becomes more reliable

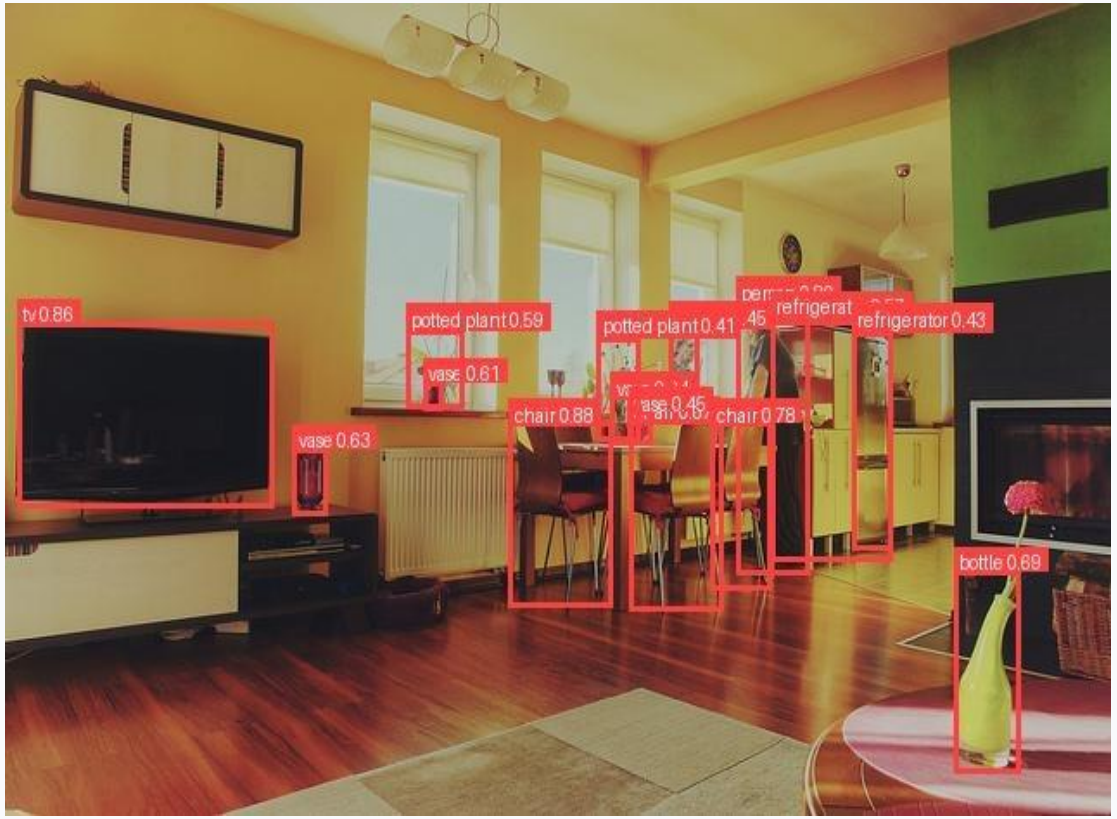
Track C: Uncertainty-Weighted Tuning

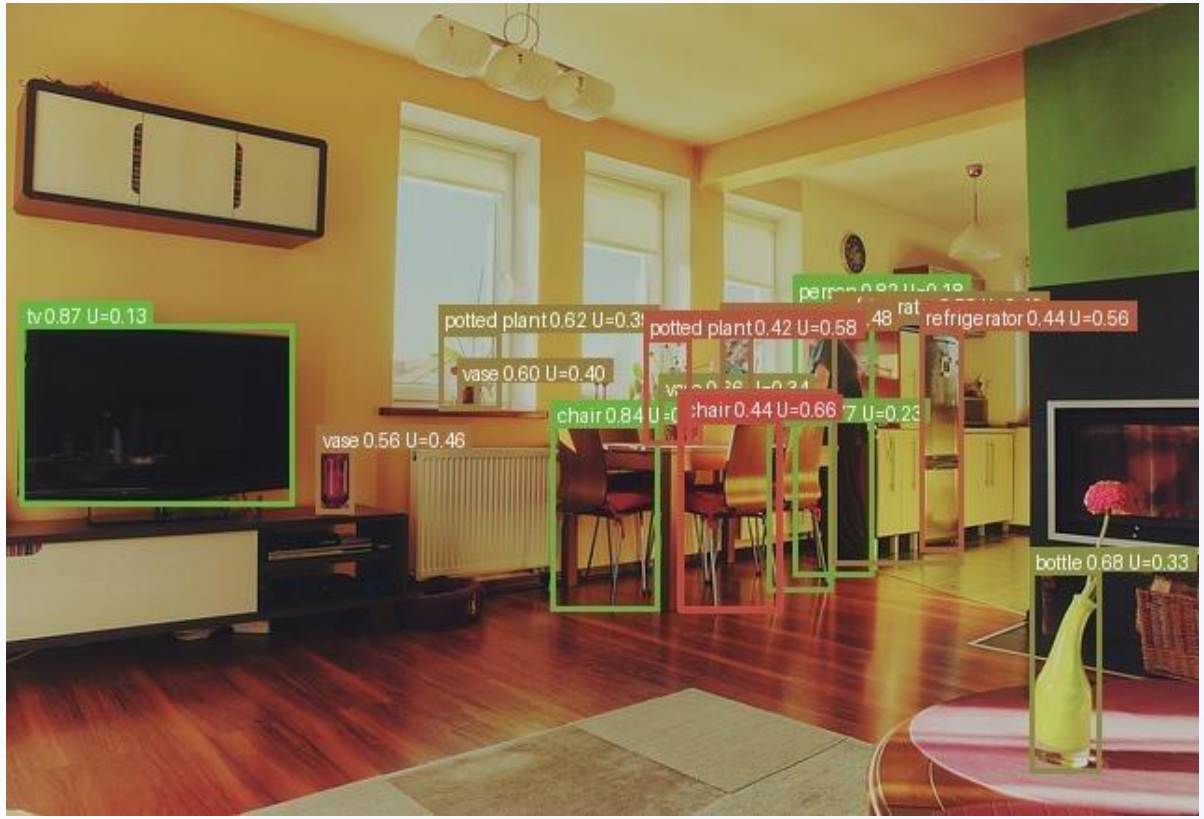
Feedback Loop: Training on Difficulty

- **Dynamic Weighting:** High uncertainty samples are given higher loss priority.
- **Small Object AP:** Focuses the optimizer on small, hard-to-detect objects.
- **Robustness:** Model becomes harder to "fool" by ambiguous inputs.
- **AP_L vs AP_S:** Better balance across object scales.

Traditional training treats all samples equally. Our approach uses Uncertainty-Weighted Learning to focus the model's optimization on challenging regions.

System Evolution: Before vs. Now





System Evolution: Before vs. Now

Baseline: Deformable DETR

- ✗ **Deterministic Points:** Predicts fixed coordinates with no error margin.
- ✗ **Pseudo-Confidence:** Score reflects presence, not trustworthiness.
- ✗ **Risk:** "Confidently Wrong" on blurry or out-of-domain data.

Our Model: Uncertainty-Aware

- ✓ **Probabilistic Boxes:** Explicitly models boundary variance (σ^2).
- ✓ **Dual Modeling:** Separates Aleatoric (data) and Epistemic (model) noise.
- ✓ **Reliability:** Calibrated scores (ECE) for safer autonomous deployment.

Comparative Performance Results

Configuration	mAP (COCO)	AP_S (Small)	ECE (Lower = Better)	Reliability
Baseline Deformable DETR	43.8	24.2	0.128	LOW
DETR + Track A (Aleatoric)	43.7	24.5	0.082	FAIR
DETR + Track B (Epistemic)	43.5	24.1	0.076	GOOD
Uncertainty-Aware Model (Full)	**43.3**	**25.1**	**0.035**	EXCELLENT

Expected Calibration Error (ECE) is reduced by ~72% compared to baseline.

Research Contributions & Impact

Technical Innovations

- Integrated probabilistic heads into DETR decoders.
- Implementation of Loss Attenuation frameworks.
- Empirical sweep of T-stochastic samples.

Practical Utility

- Real-world deployment for self-driving safety.
- Interpretable "uncertainty heatmaps" for end-users.
- Selective prediction filtering for 99.9% precision.

Team Contributions

Our team of 7 operates with an equal distribution of workload, ensuring collaborative effort across all project phases.

Maradana Kasi Sri Roshan	14.3%
Krishna Kumar Bais	14.3%
Mahathi Garapati	14.3%
Daksh Kumar Singh	14.3%
Sontam Deekshitha	14.3%
Chinthapudi Gowtham Chand	14.3%
Sevak Baliram Shekokar	14.3%

Thank You